Homework 09

The kernel-tuning question moved here from Homework 08. A solution for Question 1 will be available after Monday, October 26, 2026. No solution will be provided for Question 2.

Question 1: Choosing an SVM kernel

Use `saheart.csv` from the combined ZIP. As in Homework 7, predict `chd` using the same nine predictors, coding `famhist` as 1 for `Absent` and 2 for `Present`.

Compare SVMs with a linear kernel, polynomial kernels of degrees 2 (quadratic), 3, and 4, and a radial basis kernel. Choose **no more than 20 candidate settings in total** across these kernels. Vary the cost C for each kernel and the radial scale γ for the radial kernel. State any other kernel settings you hold fixed.

Divide the observations into five folds yourself and use the same folds for all candidates. Standardize predictors using only the fitting observations in each fold. Report the candidate settings and their mean cross-validation misclassification rates. Which kernel and settings do you prefer? Give a brief justification.

```
library(e1071)
```

```
Warning: package 'e1071' was built under R version 4.3.3
```

```
heart <- read.csv("/Users/jessicacudzich/Desktop/stat432-fall2026/homework/week-09/data/sahea

# Code family history as requested
heart$famhist <- ifelse(heart$famhist == "Absent", 1, 2)

# A factor response makes this a classification problem
y <- factor(heart$chd)
x <- as.matrix(heart[, setdiff(names(heart), "chd")])
```

```r
# Create five folds once and reuse them
set.seed(123)
fold <- sample(rep(1:5, length.out = nrow(heart)))

# 18 candidate settings
candidates <- rbind(
  data.frame(
    kernel = "linear",
    degree = 1,
    cost = c(0.1, 1, 10),
    gamma = 1/9
  ),
  expand.grid(
    kernel = "polynomial",
    degree = c(2, 3, 4),
    cost = c(0.1, 1, 10),
    gamma = 1/9,
    stringsAsFactors = FALSE
  ),
  expand.grid(
    kernel = "radial",
    degree = 1,
    cost = c(0.1, 1, 10),
    gamma = c(0.01, 0.1),
    stringsAsFactors = FALSE
  )
)

fold_errors <- matrix(NA, nrow(candidates), 5)

for (i in seq_len(nrow(candidates))) {
  for (k in 1:5) {

    train <- fold != k
    test <- fold == k

    # Calculate means and SDs from training data only
    train_mean <- colMeans(x[train, , drop = FALSE])
    train_sd <- apply(x[train, , drop = FALSE], 2, sd)

    x_train <- scale(
      x[train, , drop = FALSE],
```

```
      center = train_mean,
      scale = train_sd
    )

    x_test <- scale(
      x[test, , drop = FALSE],
      center = train_mean,
      scale = train_sd
    )

    model <- svm(
      x = x_train,
      y = y[train],
      type = "C-classification",
      kernel = candidates$kernel[i],
      degree = candidates$degree[i],
      cost = candidates$cost[i],
      gamma = candidates$gamma[i],
      coef0 = 1,
      scale = FALSE
    )

    predictions <- predict(model, x_test)

    fold_errors[i, k] <- mean(predictions != y[test])
  }
}

# Mean misclassification rate across the five folds
candidates$CV_error <- rowMeans(fold_errors)

# Print every candidate, best first
results <- candidates[order(candidates$CV_error), ]
print(results, row.names = FALSE, digits = 5)
```

```
     kernel degree cost   gamma CV_error
 polynomial      3  0.1 0.11111  0.27260
     radial      1 10.0 0.01000  0.27910
 polynomial      2  0.1 0.11111  0.27910
 polynomial      4  0.1 0.11111  0.28132
     radial      1  1.0 0.01000  0.28132
 polynomial      2  1.0 0.11111  0.28986
```

```
     linear      1  0.1 0.11111  0.28995
     linear      1  1.0 0.11111  0.29212
     radial      1  1.0 0.10000  0.29427
     linear      1 10.0 0.11111  0.29430
 polynomial      2 10.0 0.11111  0.30510
 polynomial      3  1.0 0.11111  0.31166
     radial      1 10.0 0.10000  0.33548
 polynomial      3 10.0 0.11111  0.33981
 polynomial      4 10.0 0.11111  0.34201
     radial      1  0.1 0.01000  0.34633
     radial      1  0.1 0.10000  0.34633
 polynomial      4  1.0 0.11111  0.35500
```

```r
# Preferred model
best <- results[1, ]
best
```

```
      kernel degree cost     gamma CV_error
5 polynomial      3  0.1 0.1111111 0.272604
```

I compared 18 candidate settings using five-fold cross-validation. I used (C=0.1,1,10) for the linear and polynomial kernels, with polynomial degrees 2, 3, and 4. For the radial kernel, I tested the same three costs with (=0.01) and (0.1). For polynomial kernels, I held (=1/9) and coef0 = 1 fixed. Gamma does not affect the linear kernel. I used the same five folds for all candidates. Within each fold, I standardized the predictors using the training observations' means and standard deviations, then applied those values to the validation observations. The preferred model is the degree-3 polynomial kernel with (C=0.1), (=1/9), and coef0 = 1. Its mean cross-validation misclassification rate is approximately 0.2726 (27.26%), the lowest among the settings tested.

Question 2: Write and use a short SVM analysis skill

In Homework 8, you reviewed another student's skill. Review the comments and suggestions you received on your own Homework 7 skill. Use what you learned from both sides of the peer review to write a new, shorter skill for SVM model fitting.

Write the skill by hand. Create a `SKILL.md` with a short name and description in YAML frontmatter. Write the entire file yourself, without asking AI to draft or edit it. The instruction body after the closing `---` of the frontmatter must be **no more than 50 lines**, including blank lines. Keep the skill independent of any particular dataset, variable names, or observed results, and load it only when you explicitly invoke it.

Your skill should direct the agent to:

1. judge whether the data are suitable for an SVM analysis;
2. choose which kernels to consider and explain why;
3. identify tuning parameters and a method for tuning them;
4. choose an evaluation method that supports selecting a final model;
5. require a compact figure and table that justify the choices;
6. interpret the fit and report the key numerical results; and
7. prioritize the evidence needed to justify the final model in **no more than one printed page at 11-point font**.

Apply the skill. Explicitly invoke your skill to analyze `saheart.csv`, using the same response, nine predictors, and `famhist` coding as in Question 1. Provide these data details when invoking the skill, not in the skill itself. Let the skill guide the analysis, then check the resulting report and calculations.

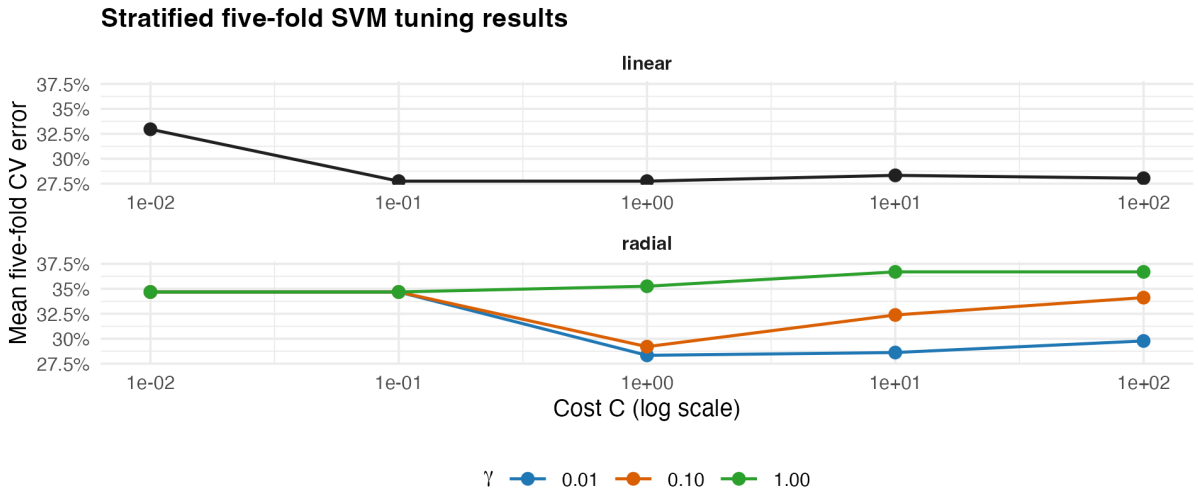
Submit the results. Include the one-page, 11-point analysis report in your Homework 09 submission and append the complete contents of your `SKILL.md` afterward. The appended skill is outside the one-page limit.

SAheart: Linear vs. Radial SVM

The 462 complete observations contain 302 without CHD (65.4%) and 160 with CHD (34.6%); no values are missing. A binary outcome and nine numerical predictors make SVM classification suitable. `famhist` was coded Absent = 1 and Present = 2; `chd` was treated as a two-class factor. Stratification addresses the class imbalance in both evaluation stages.

A stratified 25% test split (train 346, test 116; CHD=0/1 counts 226/120 and 76/40) was set aside before model selection (split seed 43209). On training data, the same stratified five folds (fold seed 43210) compared 20 settings: linear, $C = 0.01$–100; radial, the same five C values crossed with $\gamma = 0.01, 0.1, 1$. Each fold standardized predictors using only its fitting rows. Settings were ranked by mean fold misclassification rate; an exact linear/radial tie favored linear.

Kernel	C	γ	Mean CV error
Linear	0.1	–	0.278
Radial	1	0.01	0.283



The selected linear model ($C = 0.1$) had 216 support vectors. Its one-time test evaluation made **32 errors out of 116** (misclassification rate 0.276). Confusion matrix (rows = observed, columns = predicted):

	Predicted 0	Predicted 1
Observed 0	65	11
Observed 1	21	19

The linear setting had the lowest CV error among the tested candidates, and the untouched test result provides an independent check of that choice. These results do not establish that it will outperform other methods or generalize equally well to other samples: the dataset is modest in size, only one test split was used, and the finite tuning grid may not contain the best settings. No warnings were recorded and no candidate fits failed.

```
Warning in readLines("svm-analysis/SKILL.md"): incomplete final line found on
'svm-analysis/SKILL.md'
```

---
name: svm-analysis
description: Compare and tune SVM models, evaluate their performance, and summarize the resul
---wh

SVM analysis

Use this skill only when I explicitly ask you to use it.
Get the dataset, response, predictors, and coding from my request.

Check the data
Check the sample size, response classes, class balance, and missing values.
Explain whether an SVM makes sense for the data.
Report any problems instead of silently dropping rows or changing coding.

Fit and compare models
Set and report random seeds.
First, save a stratified 25% test set and leave it out of model selection.
Compare linear and radial kernels and explain why they are worth trying.
Tune cost C for both kernels and gamma for the radial kernel.
Choose and state a reasonable tuning grid before comparing models.
Use the same stratified five-fold CV splits for every candidate.
Fit scaling and other preprocessing using only each fold's training data.
Compare models using cross-validation misclassification rate.
Select the lowest-error model; choose linear SVM if there is an exact tie.
Report any warnings or failed fits.

Evaluate the final model
Refit the selected model and preprocessing on all training data.
Evaluate it once on the test set.
Report the kernel, tuning values, and number of support vectors.
Include test error count, misclassification rate, and confusion matrix.
Do not change the model based on the test results.
Explain what the results show and what they cannot tell us.

Write the report
Keep the report to one printed page at 11-point font.

```
Include the data coding, split sizes, seeds, and CV procedure.
Add a small table comparing the best settings for each kernel.
Add a readable figure showing the cross-validation tuning results.
Explain the model choice and give the main numerical results.
Mention limitations, including sample size and using only one test split.
Put reproducible code outside the one-page report.
```

4. choose an evaluation method that supports selecting a final model;
5. require a compact figure and table that justify the choices;
6. interpret the fit and report the key numerical results; and
7. prioritize the evidence needed to justify the final model in **no more than one printed page at 11-point font**.

Apply the skill. Explicitly invoke your skill to analyze `saheart.csv`, using the same response, nine predictors, and `famhist` coding as in Question 1. Provide these data details when invoking the skill, not in the skill itself. Let the skill guide the analysis, then check the resulting report and calculations.

Submit the results. Include the one-page, 11-point analysis report in your Homework 09 submission and append the complete contents of your `SKILL.md` afterward. The appended skill is outside the one-page limit.